

LEVERAGE

1. Income Statement

Particulars	Amount
Sales	---
Less: Variable Cost	---
Contribution	---
Less: Fixed Cost	---
Earnings Before Interest & Tax (EBIT)	---
Less: Interest	---
Earnings Before Tax (EBT)	---
Less: Tax	---
Earnings After Tax (EAT)	---
Less: Preference Dividend	---
Earnings Available for Equity (EAFE)	---
Number of Equity Shares	---
Earnings per Share	---

OL
 CL
 FL

2. PV Ratio = $\frac{\text{Contribution}}{\text{Sales}} \times 100 = 100 - \text{Variable Cost Ratio}$

VC Ratio = $\frac{\text{Variable Cost}}{\text{Sales}} \times 100$

3. $\text{PBT} \times (1 - t) = \text{PAT}$

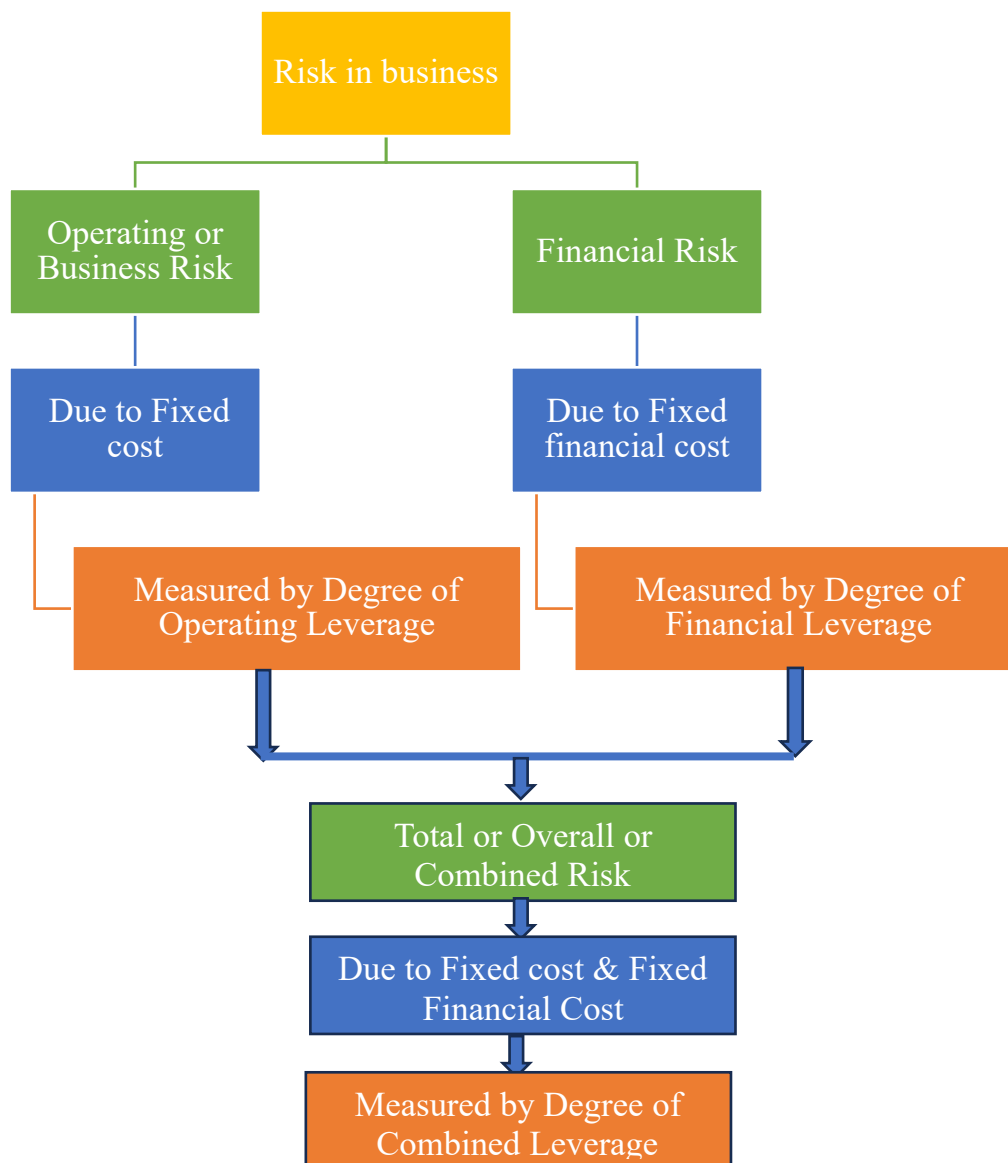
$\text{PBT} = \frac{\text{PAT}}{(1-t)}$

4. Double means increase by 100%

Triple means increase by 200%

Becomes zero means decrease by 100%

5. Risks in the business



6. Degree of Operating Leverage (DOL)

$$DOL = \frac{\% \text{ Change in EBIT}}{\% \text{ Change in Sales}}$$

$$DOL = \frac{\text{Contribution}}{EBIT}$$

$$DOL = \frac{1}{\text{Margin of Safety (MOS)}}$$

7. Degree of Financial Leverage (DFL)

$$DFL = \frac{\% \text{ Change in EPS}}{\% \text{ Change in EBIT}}$$

$$DFL \text{ (without preference shares)} = \frac{EBIT}{EBT}$$

$$DFL \text{ (with preference shares)} = \frac{EBIT}{EBT - \left(\frac{PD}{1-t}\right)}$$

8. Degree of Combined Leverage (DCL)

$$DCL = \frac{\% \text{ Change in EPS}}{\% \text{ Change in Sales}}$$

$$DCL \text{ (without preference shares)} = \frac{\text{Contribution}}{EBT}$$

$$DCL \text{ (with preference shares)} = \frac{\text{Contribution}}{EBT - \left(\frac{PD}{1-t}\right)}$$

$$DCL = DOL \times DFL$$

9. Higher the level of leverage, high will be the level of that particular risk and vice-versa.

Question – 1

A company had the following balance sheet as on 31st March, 2021:

Liabilities	₹ in crores	Assets	₹ in crores
Equity share capital (75 lakhs shares of ₹ 10 each)	7.50	Building	12.50
Reserve and Surplus	1.50	Machinery	6.25
15% Debentures	15.00	Current Assets	
Current Liabilities	6.00	Stock	3.00
		Debtors	3.25
		Bank Balance	5.00
	30.00		30.00

The additional information given is as under:

Fixed cost per annum (excluding interest) → ₹ 6 crores

Variable operating cost ratio 1 PV Ratio = 40%. → 60%

Total assets turnover ratio → 2.5

Income tax rate → 40%

Calculate the following and comment: $\frac{13.05 \text{ cr.}}{(7.5 \div 0.95)} = 17.40$

- Earnings per share
- Operating leverage → $30 \div 24 =$
- Financial leverage → $24 \div 21.75 =$
- Combined leverage → $30 \div 21.75 =$

$TA \text{ turn. ratio} = \frac{\text{Sales}}{TA}$
 $d.s = \frac{\text{Sales}}{30 \text{ cr.}}$
 $\text{Sales} = 75 \text{ cr.}$
 $\rightarrow VC @ 60\% = 45$
 $Contribution = 30$
 $\rightarrow FC = 6$
 $EBIT = 24$
 $\rightarrow Int. = 2.25$
 $(15 \times 15\%)$
 $EBT = 21.75$
 $\rightarrow Tax = 8.7$
 $EAT = 13.05$

Solution

$$\text{Total assets turnover ratio} = \frac{\text{Sales}}{\text{Total Assets}}$$

$$2.5 = \frac{\text{Sales}}{30 \text{ crores}}$$

$$\text{Sales} = ₹ 75 \text{ Crores}$$

Income Statement

<u>Particulars</u>	<u>Amount (₹)</u>
Sales	75,00,00,000
Less: Variable Cost @ 60%	45,00,00,000
Contribution	30,00,00,000
Less: Fixed Cost	6,00,00,000
EBIT	24,00,00,000
Less: Interest (15 crore × 15%)	2,25,00,000
EBT	21,75,00,000
Less: Income tax @ 40%	8,70,00,000
EAT/EAE	13,05,00,000

(a) Earning per share = $\frac{EAE}{\text{No. of equity shares}} = \frac{13,05,00,000}{75,00,000} = \underline{\underline{₹ 17.40 \text{ per share}}}$

It indicates the amount the company earns per share. It is used as a guide for valuing the share and making investment decisions by the investor.

(b) Operating Leverage = $\frac{\text{Contribution}}{EBIT} = \frac{30,00,00,000}{24,00,00,000} = \underline{\underline{1.25 \text{ times}}}$

It indicates the structure of fixed cost in the business. It indicates sensitivity of earnings before interest and tax (EBIT) to changes in sales at a particular level.

(c) Financial Leverage = $\frac{EBIT}{EBT} = \frac{24,00,00,000}{21,75,00,000} = \underline{\underline{1.103 \text{ times}}}$

It indicates the use of fixed financial cost in the capital structure. It indicates sensitivity of earning per share (EPS) to changes in earnings before interest and tax (EBIT) at a particular level.

(d) Combined Leverage = $OL \times FL = \underline{\underline{1.25 \times 1.103 = 1.37875}}$

It indicates the choice of fixed cost and fixed financial cost in the capital structure used. It indicates the sensitivity of earning per share (EPS) to changes in sales at a particular level.

Question – 2

Calculate the operating leverage, financial leverage and combined leverage from the following data under Situation I and II and Financial Plan A and B:

Installed Capacity	→ 4,000 units	} Prod. = 4000 × 75% = 3000	
Actual Production and Sales	→ 75% of capacity		
Selling Price	→ ₹ 30 per unit	} Cont. Pr. = 15	
Variable Cost	→ ₹ 15 per unit		
Fixed Cost:		(I)	(II)
Under Situation I	→ ₹ 15,000	Cont. 45000	45000
Under Situation II	→ ₹ 20,000	FC 15000	20000
		EBIT 30000	25000
		OL	

	Financial Plan	
	A (₹)	B (₹)
Equity	→ 10,000	→ 15,000
Debt (Rate of interest at <u>20%</u>)	→ ✓ 10,000	→ ✓ 5,000
	20,000	20,000

Solution

Particulars	Situation I	Situation II
Contribution [$4,000 \times 75\% \times (30 - 15)$]	45,000	45,000
Less: fixed cost	(15,000)	(20,000)
EBIT	✓ 30,000	✓ 25,000
Operating Leverage (Contribution/EBIT)	1.5	1.8

Calculation of Financial & Combined Leverage

Financial Plan	Situation I		Situation II	
	A	B	A	B
EBIT	✓ 30,000	✓ 30,000	→ 25,000	→ 25,000
Less: Interest on debt	✓ (2,000)	✓ (1,000)	✓ (2,000)	✓ (1,000)
EBT	28,000	29,000	23,000	24,000
Financial Leverage (EBIT/EBT)	1.07	1.03	1.09	1.04

Question – 3

Following is the Balance Sheet of Gitashree Ltd. is given below:

Liabilities	Amount (₹)
Shareholder's Fund	
Equity Share Capital (₹ 10 each) → $\frac{1.800}{10} = 18000$	→ ✓ 1,80,000
Reserve & Surplus	→ 60,000
Non-Current Liabilities (10% Debentures)	→ ✓ 2,40,000
Current Liabilities	→ 1,20,000
Total	6,00,000
Non-Current Assets	4,50,000
Current Assets	1,50,000
Total	6,00,000

The company's total assets turnover ratio is 4. Its fixed operating cost is ₹ 2,00,000 and its variable operating cost ratio is 60%. The income tax rate is 30%. Calculate:

- (a) Degree of operating leverage → $9.602 \div 7.602 =$
 (b) Degree of financial leverage → $7.602 \div 7.362 =$
 (c) Degree of combined leverage → $9.602 \div 7.362 =$
- Find out EBIT if EPS is (a) ₹ 1; (b) ₹ 2; and (c) ₹ 0.

$$EPS = \frac{(EBIT - Int.) (1 - t) - Pref. Div.}{No. of Eq. Sh.} \Rightarrow 1 = \frac{(EBIT - 24000)(1 - 0.30) - 0}{18000}$$

$$\Rightarrow 18000 = (0.7)(EBIT) - 16800$$

$$\Rightarrow EBIT = \frac{34800}{0.7} = 49714$$

Solution

$$\text{Total assets turnover ratio} = \frac{\text{Sales}}{\text{Total Assets}}$$

$$4 = \frac{\text{Sales}}{6,00,000}$$

$$\text{Sales} = ₹ 24,00,000$$

Income Statement

<u>Particulars</u>	<u>Amount (₹)</u>
Sales	24,00,000
Less: Variable Cost @ 60%	<u>14,40,000</u>
Contribution	9,60,000
Less: Fixed Cost	<u>2,00,000</u>
EBIT	7,60,000
Less: Interest (2,40,000 × 10%)	<u>24,000</u>
EBT	7,36,000
Less: Income tax @ 30%	<u>2,20,800</u>
EAT/EAE	<u>5,15,200</u>

$$(1) \quad (a) \quad \text{Operating Leverage} = \frac{\text{Contribution}}{\text{EBIT}} = \frac{9,60,000}{7,60,000} = 1.263 \text{ times}$$

$$(b) \quad \text{Financial Leverage} = \frac{\text{EBIT}}{\text{EBT}} = \frac{7,60,000}{7,36,000} = 1.033 \text{ times}$$

$$(c) \quad \text{Combined Leverage} = \text{OL} \times \text{FL} = 1.263 \times 1.033 = 1.304 \text{ times}$$

$$(2) \quad (a) \quad \text{EPS} = \frac{(\text{EBIT} - \text{Interest})(1-t)}{\text{No. of equity shares}}$$

$$1 = \frac{(\text{EBIT} - 24,000)(1-0.30)}{18,000}$$

$$\text{EBIT} = ₹ 49,714$$

$$(b) \quad \text{EPS} = \frac{(\text{EBIT} - \text{Interest})(1-t)}{\text{No. of equity shares}}$$

$$2 = \frac{(\text{EBIT} - 24,000)(1-0.30)}{18,000}$$

$$\text{EBIT} = ₹ 75,429$$

$$(c) \quad \text{EPS} = \frac{(\text{EBIT} - \text{Interest})(1-t)}{\text{No. of equity shares}}$$

$$0 = \frac{(\text{EBIT} - 24,000)(1-0.30)}{18,000}$$

$$\text{EBIT} = ₹ 24,000$$

$$FL = \frac{EBIT}{EBT} \Rightarrow 1.39 = \frac{13.50}{EBT} \Rightarrow EBT = 971223$$

Question – 4

The information related to XYZ Company Ltd. for the year ended 31st March, 2020 are as follows:

Equity Share capital of ₹ 100 each	→ ₹ 50 lakhs
✓ 12% Bonds of ₹ 1,000 each	→ ₹ 30 lakhs
Sales	→ ₹ 84 lakhs
Fixed cost (excluding interest)	→ ₹ 7.50 lakhs
Financial leverage	1.39
Profit-volume ratio	→ 25% ✓
Market price per equity share	→ ₹ 200
Income tax applicable	→ 30%

Sales	840
(-) VC	
Cont. @ 25%	210
(-) FC	7.500
EBIT	13.500
(-) Bond Int.	3.600
(-) other (diff)	18775
EBT	971223
(-) Tax @ 30%	291367
EAT/EAT	679856

You are required to calculate:

- (a) Operating Leverage → $210 \div 13.50 =$
- (b) Combined Leverage → $210 \div 971223 =$
- (c) Earnings per share → $679856 \div 50000 = 13.597$
- (d) Earning Yield

$$\frac{EPS}{MPS} \times 100 = \frac{13.597}{200} \times 100 =$$

Solution**Income Statement**

Particulars	Amount (₹)
Sales	84,00,000
Less: Variable cost (84,00,000 × 75%)	63,00,000
Contribution (84,00,000 × 25%)	21,00,000
Less: Fixed cost	7,50,000
EBIT	13,50,000
Less: Interest on bonds (12% × 30 lakhs)	3,60,000
Less: Other fixed interest (bal. figure)	18,777
EBT (13,50,000 ÷ 1.39)	9,71,223
Less: Tax @ 30%	2,91,367
EAT	6,79,856

(a) Operating Leverage = $\frac{\text{Contribution}}{\text{EBIT}} = \frac{21,00,000}{13,50,000} = 1.56 \text{ times}$

(b) Combined Leverage = Operating Leverage × Financial Leverage = $1.56 \times 1.39 = 2.13$

(c) Earnings per share (EPS) = $\frac{\text{EAT}}{\text{No. of shares outstanding}} = \frac{6,79,856}{50,000} = ₹ 13.597$

(d) Earning yield = $\frac{\text{EPS}}{\text{Market price per share}} \times 100 = \frac{13.597}{200} \times 100 = 6.798\%$

$$FL = \frac{EBIT}{EBIT - \text{Int.}} \Rightarrow 4 = \frac{y}{y - 3000} \Rightarrow 4y - 12000 = y \Rightarrow y = \frac{12000}{3} = 4000$$

$$\frac{20000}{25\%}$$

Question – 5

From the following information, prepare Income Statement of Company A & B:

$$OL = \frac{\text{Cont. EBIT}}{\text{Cont. EBIT} - \text{Int.}}$$

$$S = \frac{\text{Cont. EBIT}}{\text{Cont. EBIT} - \text{Int.}}$$

$$\text{Cont.} = 20000$$

Particulars	Company A	Company B
Margin of safety	0.20	0.25
Interest	₹ 3,000	₹ 2,000
Profit volume ratio	0.25%	33.33%
Financial Leverage	4	3
Tax rate	45%	45%

Sales	80000
(-) VC	60000
Cont.	20000
(-) FC (Net)	16000
EBIT	4000
(-) Int.	3000
EBT	1000
(-) Tax	450
EAT	550

Solution

Income Statement

Particulars	Company A (₹)	Company B (₹)
Sales	80,000	36,000
(-) Variable cost	60,000	24,000
Contribution	20,000	12,000
(-) Fixed cost	16,000	9,000
EBIT	4,000	3,000
(-) Interest	3,000	2,000
EBT	1,000	1,000
(-) Tax @ 45%	450	450
EAT	550	550

Working Notes:

(i) Company A

Financial leverage

4 times

4(EBIT) - ₹ 12,000

EBIT

$$= \frac{EBIT}{EBIT - \text{Interest}}$$

$$= \frac{EBIT}{EBIT - 3,000}$$

$$= EBIT$$

$$= ₹ 4,000$$

Company B

Financial leverage

3 times

3(EBIT) - ₹ 6,000

EBIT

$$= \frac{EBIT}{EBIT - \text{Interest}}$$

$$= \frac{EBIT}{EBIT - 2,000}$$

$$= EBIT$$

$$= ₹ 3,000$$

(ii) Company A

Operating leverage = $\frac{1}{\text{Margin of Safety}}$ = $1 \div 0.20$ = 5 times

Operating leverage = $\frac{\text{contribution}}{EBIT}$

5 times = $\frac{\text{contribution}}{4,000}$

Contribution = ₹ 20,000

Company B

$$\text{Operating leverage} = \frac{1}{\text{Margin of Safety}} = 1 \div 0.25 = 4 \text{ times}$$

$$\text{Operating leverage} = \frac{\text{Contribution}}{\text{EBIT}}$$

$$4 \text{ times} = \frac{\text{Contribution}}{3,000}$$

Contribution = ₹ 12,000

(iii) Company A

$$\text{Sales} = \frac{\text{Contribution}}{\text{PV Ratio}} = \frac{20,000}{25\%} = ₹ 80,000$$

Company B

$$\text{Sales} = \frac{\text{Contribution}}{\text{PV Ratio}} = \frac{12,000}{33.33\%} = ₹ 36,000$$

$$OL = \frac{\text{Cont.}}{\text{EBIT}} \Rightarrow 2.5 = \frac{100}{\text{EBIT}} \Rightarrow \text{EBIT} = 40$$

$$CL = \frac{\% \Delta \text{ in EPS}}{\% \Delta \text{ in Sales}} = \frac{100\%}{25\%} = 4$$

$$\frac{\text{Cont.}}{\text{EBIT} - \text{Int.}} = 4 \Rightarrow \frac{100}{(40 - \text{Int.})} = 4 \Rightarrow 40 - \text{Int.} = 2.50$$

$$\text{Debt} = 1.50 \div 16\% = 937500$$

$$\text{EPS} = \frac{(40 - 1.50)(1 - 0.50)}{12} = 1.25$$

Question – 6

The sales revenue of SK Ltd. @ ₹ 20 per unit of output is ₹ 20 lakhs and contribution is ₹ 10 lakhs. At the present level of output, the DOL of the company is 2.5. the company does not have any Preference Shares. The number of Equity Shares are 1 lakh. Applicable corporate income tax rate is 50% and the rate of interest on Debt Capital is 16% p.a. Calculate the Eps (at sales revenue of ₹ 20 lakhs) and amount of debt capital of the company if a 25% decline in Sales will wipe out EPS.

Solution

(i) Calculation of Fixed Cost

$$\text{DOL} = \frac{\text{Contribution}}{\text{Contribution-Fixed Cost}} \text{ or } 2.5 = \frac{₹10,00,000}{\text{EBIT}} \text{ or EBIT} = ₹4,00,000$$

$$\text{EBIT} = \text{Contribution} - \text{Fixed Cost}$$

$$₹4,00,000 = ₹10,00,000 - \text{Fixed Cost}$$

$$\text{Fixed Cost} = ₹10,00,000 - ₹4,00,000 = ₹6,00,000$$

(ii) Calculation of Degree of Combined Leverage (DCL)

Question says that 25% change in sales will wipe out EPS. Here, wipe out means it will reduce EPS by 100%.

$$\text{DCL} = \frac{\text{Percentage Change in EPS}}{\text{Percentage Change in Sales}} = \frac{100\%}{25\%} = 4$$

(iii) Calculation of Degree of Financial Leverage (DFL)

$$\text{DCL} = \text{DOL} \times \text{DFL}$$

$$4 = 2.5 \times \text{DFL}$$

$$\text{So, DFL} = 1.6$$

(iv) Calculation of Interest and amount of Debt

$$DFL = \frac{EBIT}{EBIT - \text{Int}} \text{ Or, } 1.6 = \frac{₹4,00,000}{₹4,00,000 - \text{Int}} \text{ Or, Int} = ₹1,50,000$$

Debt × Interest rate = Amount of Interest

$$\text{Debt} \times 16\% = ₹1,50,000$$

$$\text{Debt} = ₹9,37,500$$

(v) Calculation of Earnings per share (EPS)

$$EPS = \frac{(EBIT - \text{Int})(1 - t)}{N} = \frac{(₹4,00,000 - ₹1,50,000)0.5}{1,00,000} = ₹1.25$$

Question – 7

Information of A Ltd. is given below:

- Earnings after tax: 5% on sales ✓
- Income tax rate: 50% ✓
- Degree of operating leverage: 4 times ✓
- 10% Debenture in capital structure: ₹ 3 lakhs ✓
- Variable costs: ₹ 6 lakhs ✓

Required:

(i) From the given data complete the following statement:

Sales	y	XXXX	120
Less: Variable costs		6,00,000	60
Contribution	y - 60	XXXX	60
Less: Fixed costs		XXXX	4.50
EBIT		XXXX	1.50 ✓
Less: Interest expenses	30000	XXXX	0.30
EBT		XXXX	1.20 -
Less: Income tax		XXXX	0.60
EAT	5% × y = 0.05y	XXXX	0.60

(ii) Calculate Financial Leverage and Combined Leverage.

(iii) Calculate the percentage change in earnings per share, if sales increased by 5%.

Solution

Let sales = y

$$\text{Degree of operating leverage} = \frac{\text{Contribution}}{EBIT}$$

$$4 = \frac{\text{Contribution}}{EBIT}$$

$$4(EBIT) = \text{Sales} - \text{Variable cost}$$

$$4(EBIT) = \text{Sales} - 6,00,000$$

$$EBIT = 0.25(y) - 1,50,000 \dots \dots \dots (i)$$

Also, given Earning after tax = 5% of sales

$$5\% \times \text{Sales} = (EBIT - \text{Interest})(1 - t)$$

$$0.05y = [0.25y - 1,50,000 - (3,00,000 \times 10\%)](1 - 0.50)$$

$$0.05y = (0.25y - 1,80,000)(0.50)$$

$$4 = \frac{y - 60}{EBIT} \Rightarrow EBIT = \frac{y - 60}{4} = (0.25)y - 1.50$$

$$EAT = (EBIT - \text{Int.}) (1 - t)$$

$$0.05y = (0.25y - 1.50 - 0.30)(1 - 0.50)$$

$$0.05y = (0.25y - 1.80)(0.5)$$

$$0.05y = 0.125y - 90000$$

$$y = \frac{90000}{0.075} = 120$$

$$EBIT = (0.25)(120) - 1.50 = 1.50$$

$$CL = \frac{\% \Delta \text{ in EPS}}{\% \Delta \text{ in Sales}}$$

$$S = \frac{\% \Delta \text{ EPS}}{5\%}$$

$$\% \Delta \text{ EPS} = 25\%$$

$$0.05y = 0.125y - 90,000$$

$$0.075y = 90,000$$

$$y = 12,00,000$$

$$\text{Thus, EBIT} = 0.25(12,00,000) - 1,50,000 = 1,50,000$$

$$\text{Fixed cost} = \text{Contribution} - \text{EBIT} = (12,00,000 - 6,00,000) - 1,50,000 = 4,50,000$$

Income Statement

Sales	12,00,000
Less: Variable costs	6,00,000
Contribution	6,00,000
Less: Fixed costs	4,50,000
EBIT	1,50,000
Less: Interest expenses (3,00,000×10%)	30,000
EBT	1,20,000
Less: Income tax @50%	60,000
EAT	60,000

$$(a) \text{ Financial Leverage} = \frac{\text{EBIT}}{\text{EBT}} = \frac{1,50,000}{1,20,000} = 1.25 \text{ times}$$

$$\text{Combined Leverage} = \frac{\text{Contribution}}{\text{EBT}} = \frac{6,00,000}{1,20,000} = 5 \text{ times}$$

$$(b) \text{ Combined Leverage} = \frac{\% \text{ Change in EPS}}{\% \text{ Change in Sales}}$$

$$5 = \frac{\% \text{ Change in EPS}}{+5}$$

$$\% \text{ change in EPS} = +25\%$$

Thus, EPS increases by 25.

$$4 = \frac{\text{EBIT}}{\text{EBT} - \text{Int.} \left(\frac{PD}{1-P} \right)}$$

$$4 = \frac{y}{y - 57400 - \left(\frac{0.15y}{1-0.2} \right)} \Rightarrow 4 = \frac{y}{y - 57400 - (0.21875)y}$$

Question – 8

From the following information for financial year 2023-24,

Financial Leverage (FL) = 4

P/V Ratio = 40%

Tax rate = 30%

Depreciation (part of manufacturing overheads) = ₹10,000

✓ Preference dividend is 15% of Operating Profit

Cash Breakeven Sales = ₹2,25,000

Equity Share Capital = ₹1,00,000

Reserves & Surplus as on 31.03.2023 = ₹35637

$$4y - 229600 - 0.8572y = y$$

$$y = \frac{229600}{2.1428}$$

$$y = 108595$$

$$\text{Cash BEP} = \frac{\text{Cash FC}}{PV} \Rightarrow 2.25 = \frac{\text{Cash FC}}{40\%}$$

$$\text{Cash FC} = 90,000$$

$$\text{Sh. fund} = 12 + 35637 + 19547 = 155184$$

Particulars	Amount (₹)
Sales (208595 ÷ 40%)	521488
(-) Variable Cost	???
Contribution	208595
(-) Fixed cost (90000 + 10000)	100000
EBIT	108595
(-) Interest expense	57,400

EBT	51195	???
(-) Tax	15359	???
EAT	35836	???
(-) Preference Dividend (15% × 108595)	16289	???
Earnings for equity shareholders	19547	???

CALCULATE –

i. Complete the Income statement for FY 2023-24

ii. Operating Leverage & Combined Leverage

iii. Percentage change in EPS, if sales increase and decreases by 7%

iv. Calculate Return on Equity Shareholders funds on 31.03.24

v. Amount of Debt, if post tax interest rate is 6.65%

$$\rightarrow \frac{57400}{9.50\%} = 604211$$

$$\rightarrow \text{Post tax rate} = \frac{6.65}{(1-0.30)} = 9.50\%$$

$$\frac{19547}{155184} \times 100 = 12.596\%$$

Solution

WN1 – Calculation of EBIT using Financial Leverage

$$FL = \frac{EBIT}{(EBIT - \text{Int.}) - \frac{D_p}{1-t}}$$

Let EBIT be 'X'

$$\text{Preference Dividend (Dp)} = 0.15X$$

$$4 = \frac{X}{(X - 57,400) - \frac{0.15X}{1-0.3}}$$

$$3.16X - 2,29,600 = X$$

$$X = \text{EBIT} = ₹ 1,06,296$$

$$\text{Preference Dividend} = 1,06,296 \times 0.15 = ₹ 15,944$$

WN2 – Calculation of Total Fixed Cost

$$\begin{aligned} \text{Cash Fixed Cost} &= \text{Cash Breakeven sales} \times \text{PV Ratio} \\ &= 2,25,000 \times 0.40 \\ &= ₹ 90,000 \end{aligned}$$

$$\begin{aligned} \text{Total Fixed Cost} &= 90,000 + 10,000 \text{ (Depreciation)} \\ &= ₹ 1,00,000 \end{aligned}$$

$$\text{Therefore Contribution} = 1,06,296 + 1,00,000 = 2,06,296$$

i. **Income Statement for FY 2023-24:**

Particulars	Amount (₹)
Sales (2,06,296 / 0.40)	5,15,740
(-) Variable Cost	(3,09,444)
Contribution	2,06,296

(-) Fixed Cost (See Wn 2)	(1,00,000)
EBIT (See Wn 1)	1,06,296
(-) Interest Exp	(57,400)
EBT	48,896
(-) Tax	(14,669)
EAT	34,227
(-) Preference Dividend (1,06,296 X 15%)	(15,944)
Earnings For Equity Shareholders	18,283

ii. **Operating Leverage** = $\frac{\text{Contribution}}{\text{EBIT}}$

$$= 2,06,296 / 1,06,296$$

$$= 1.94 \text{ times}$$

Combined Leverage = OL x FL

$$= 4 \times 1.94$$

$$= 7.76 \text{ times}$$

- iii. **Combined leverage** is the study from Sales to Eps, where combined leverage measures the % change in EPS due to increase/decrease in % change in sales

$$CL = \% \text{ Change in EPS} / \% \text{ Change in sales}$$

$$7.76 = \% \text{ Change in EPS} / 7$$

Therefore, % Change in EPS = 54.32%

If sales increase by 7%, then EPS will rise by 54.32% and on the contrary if sales decrease by 7%, then EPS will fall by 54.32%.

- iv. **Return on Equity shareholders' funds**

$$= \frac{\text{Earnings for Equity holders}}{\text{Total Eq. Shareholders' funds}}$$

$$= 18,283 / 1,53,920$$

$$= 11.88\%$$

- v. **Calculation of Amount of Debt**

$$\text{Pre-tax Interest \%} = \frac{\text{Interest rate (post tax)}}{1 - t}$$

$$= 6.65 / 0.70$$

$$= 9.5\%$$

$$\text{Debt Amount} = \frac{\text{Interest (Rs.)}}{\text{Interest (\%)}}$$

$$= 57,400 / 0.095$$

$$= ₹ 6,04,210$$

10. Relation between BEP, FC and DOL

<u>DOL</u>		<u>Fixed Cost</u>		<u>Operating BEP</u>		<u>MOS</u>
High	→	High	→	High	→	Low
Low	→	Low	→	Low	→	High

11. DOL Analysis

Situation	Result or Interpretation
No Fixed Cost	- DOL = 1 - No operating risk
High Fixed Cost	- High BEP - High DOL
Low Fixed Cost	- Low BEP - Low DOL
Sales > Operating BEP i.e. EBIT > 0	- Existing Profit - DOL is in favour
Sales < Operating BEP i.e. EBIT < 0	- Existing Loss - DOL is against

12. Financial BEP

Level of EBIT at which EPS is zero

$$\text{Financial BEP} = \text{Interest} + \frac{PD}{(1-t)}$$

$$\text{EPS} = \frac{(EBIT - \text{Interest})(1-t) - \text{Pref. Dividend}}{\text{No. of Equity Shares}}$$

13. Analysis of DFL

Situation	Result or Interpretation
No Fixed Finance Cost	- DFL = 1 - No financial risk
High Fixed Financial Cost	- High Financial BEP - High DFL

Low Fixed Financial Cost	• Low Financial BEP • Low DFL
EBIT > Financial BEP i.e. EPS > 0	• EPS is positive • DFL is in favour
EBIT < Financial BEP i.e. EPS < 0	• EPS is negative • DFL is against

14. Financial Leverage as a Double Edge Sword

For Debt:

If Rate of Interest < Return on Investment (ROI) Favourable

If Rate of Interest > Return on Investment (ROI) Unfavourable

If Rate of Interest = Return on Investment (ROI) Indifferent

For Preference:

If Rate of Preference Dividend < ROI(1 - t) Favourable

If Rate of Preference Dividend > ROI(1 - t) Unfavourable

If Rate of Preference Dividend = ROI(1 - t) Indifferent

15. Trading on Equity

When the quantity of fixed cost fund is relatively high in comparison to equity capital it is said that the firm is “trading on equity”.

16. Analysis of DCL

DOL	DFL	Result or Interpretation
High	High	• High risky situation
Low	Low	• Low risk situation
High	Low	• Moderate risk • EBIT is low • No benefit of trading on equity
Low	High	• Moderate risk • EBIT is high → ROI → High • Benefit of trading on equity available

Question – 9

Details of a company for the year ended 31st March, 2022 are given below:

Sales	→	₹ 86 lakhs
Profit Volume (P/V) Ratio	→	35%
Fixed cost excluding interest expenses	→	₹ 10 lakhs
10% Debt	→	₹ 55 lakhs
Equity Share Capital of ₹ 10 each	→	₹ 75 lakhs
Income Tax rate	→	40%

Sale	86l
Cont. @ 35%	20.1
FC	10
EBIT	20.1
Int.	5.5
EBT	14.6
Tax	5.84
EAT	8.760

Required:

- Determine company's return on capital employed (pre-tax) and Eps.
- Does the company have a favourable financial leverage? → $ROI (15.46\%) > Int. (10\%)$ Yes fav.
- Calculate operating, financial and combine leverages of the company
- Calculate percentage change in EBIT, if sales increases by 10%. $\% \Delta EBIT = 10 \times OL = \checkmark$
- At what level of sales, the Earning before Tax (EBT) of the company will be equal to zero?
- Calculate percentage change in EBT if EBIT increases by 10%. $FL = \frac{\% \Delta \text{in EBT}}{\% \Delta \text{in EBIT}} \Rightarrow \% \Delta \text{in EBT} = 10 \times FL = \checkmark$

Solution

Income Statement

Particulars	Amount (₹)
Sales	86,00,000
Less: Variable cost (86,00,000 × 65%)	55,90,000
Contribution	30,10,000
Less: Fixed cost	10,00,000
EBIT	20,10,000
Less: Interest (10% × 55,00,000)	5,50,000
EBT	14,60,000
Less: Tax @ 40%	5,84,000
EAT/EAE	8,76,000

- Return on capital employed = $\frac{EBIT}{\text{Capital employed}} \times 100 = \frac{20,10,000}{1,30,00,000} \times 100 = 15.46\%$
 Earning per share = $\frac{EAE}{\text{No. of Equity Shares}} = \frac{8,76,000}{7,50,000} = ₹ 1.168$
- Since, the return on capital employed (15.46%) is more than the interest rate (10%), thus the company has a favourable financial leverage.
- Operating leverage = $\frac{\text{Contribution}}{EBIT} = \frac{30,10,000}{20,10,000} = 1.498$ times
 Financial leverage = $\frac{EBIT}{EBT} = \frac{20,10,000}{14,60,000} = 1.377$ times
 Combined leverage = $\frac{\text{Contribution}}{EBT} = \frac{30,10,000}{14,60,000} = 2.062$ times
- Operating leverage = $\frac{\% \text{ Change in EBIT}}{\% \text{ Change in Sales}}$
 $1.498 = \frac{\% \text{ Change in EBIT}}{+10}$
 $\% \text{ Change in EBIT} = +14.98$

Thus, EBIT increases by 14.98%

$$(v) \quad \text{Required sales} = \frac{\text{Fixed cost} + \text{Interest}}{\text{PV Ratio}} = \frac{(10,00,000 + 5,50,000)}{35\%} = ₹ 44,28,571$$

$$(vi) \quad \text{Financial leverage} = \frac{\% \text{ Change in EBT}}{\% \text{ Change in EBIT}}$$

$$1.377 = \frac{\% \text{ Change in EBT}}{+10}$$

$$\% \text{ change in EBT} = 13.77\%$$